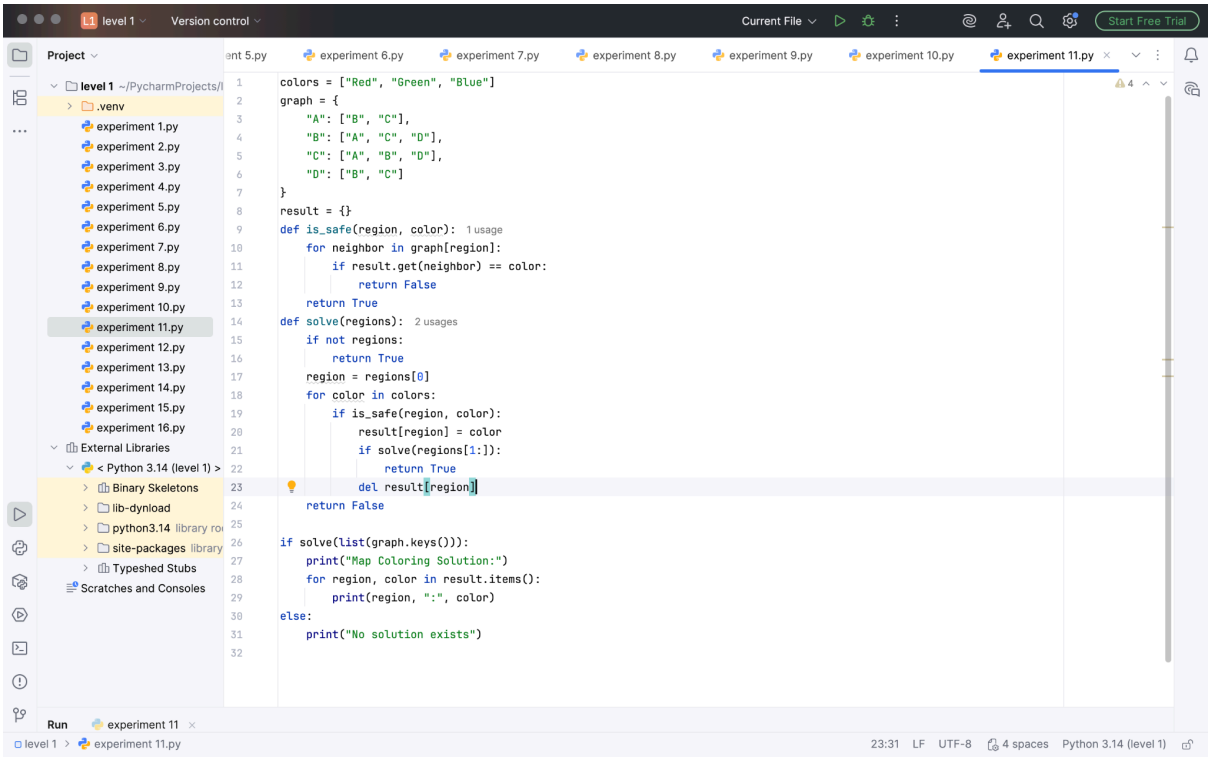
PROGRAM 11: Python program for Map Coloring to implement CSP.

Aim: To implement the Map Coloring problem using a Constraint Satisfaction Problem (CSP).

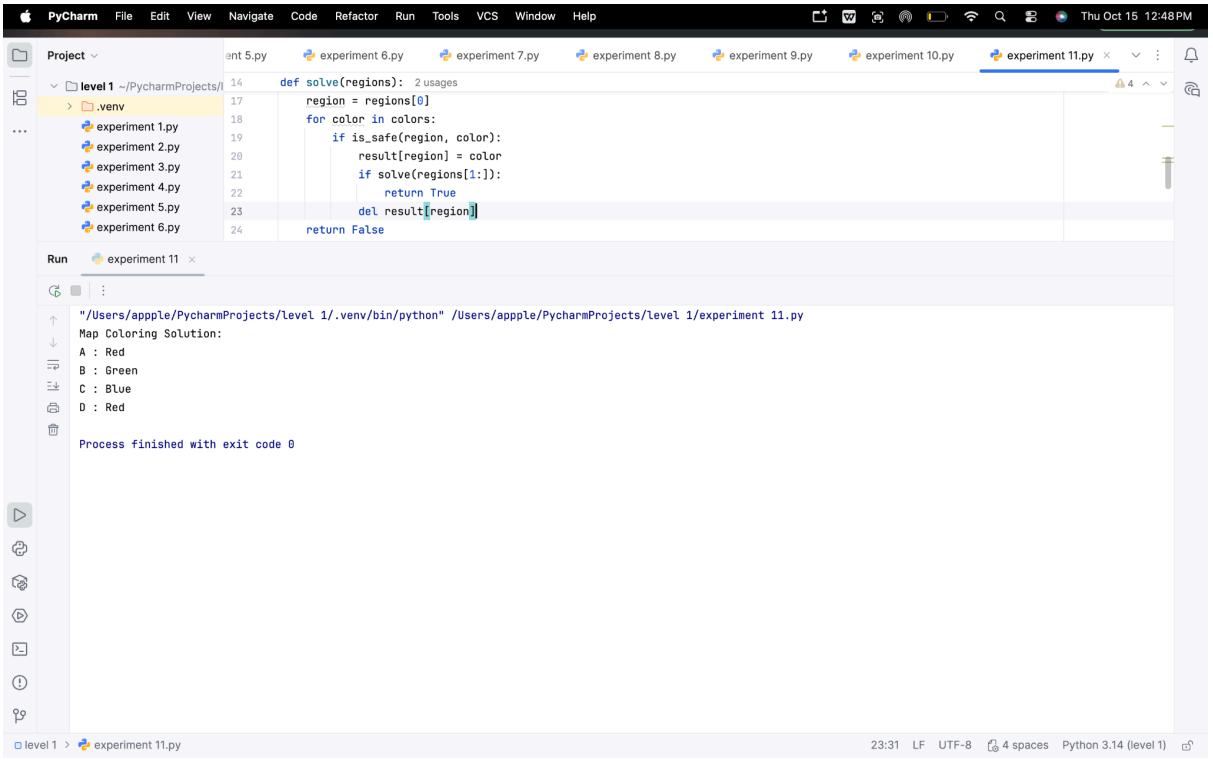
Algorithm:

1. Define the regions and their neighboring regions.
2. Assign a set of available colors.
3. Select an uncolored region.
4. Assign a color that differs from its neighbors' colors.
5. Repeat until all regions are colored.

CODE:



OUTPUT:



Result: Thus, the Map Coloring problem was implemented successfully using CSP and backtracking.

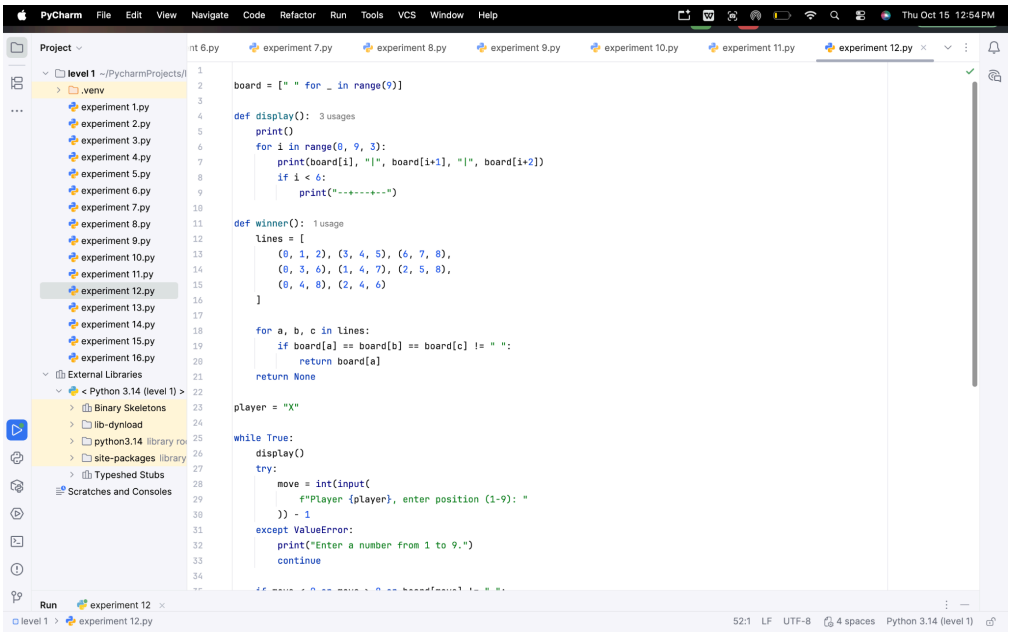
PROGRAM 12:The Python program for the Tic-Tac-Toe game

AIM: To implement a Tic-Tac-Toe game using Python.

Algorithm:

1. Create a 3 × 3 game board.
2. Allow two players to enter their moves.
3. Check whether a player has won.
4. Check for a draw if the board is full.
5. Continue until the game ends.

CODE:



```
board = [" " for _ in range(9)]

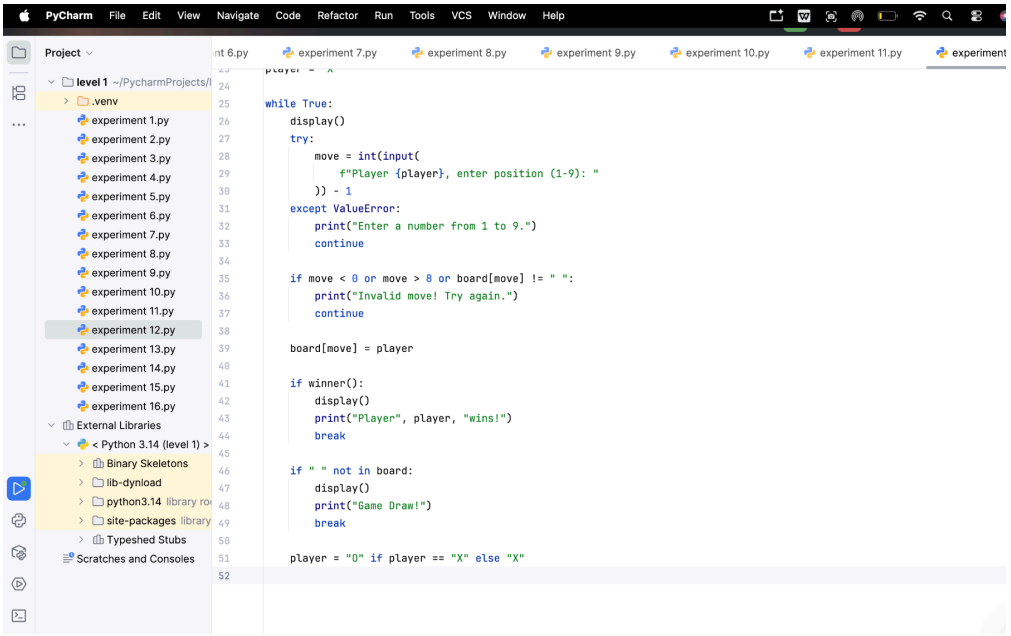
def display():
    print()
    for i in range(0, 9, 3):
        print(board[i], "|", board[i+1], "|", board[i+2])
        if i < 6:
            print("--+---+--")

def winner():
    lines = [
        (0, 1, 2), (3, 4, 5), (6, 7, 8),
        (0, 3, 6), (1, 4, 7), (2, 5, 8),
        (0, 4, 8), (2, 4, 6)
    ]

    for a, b, c in lines:
        if board[a] == board[b] == board[c] != " ":
            return board[a]
    return None

player = "X"

while True:
    display()
    try:
        move = int(input(
            f"Player {player}, enter position (1-9): "
        )) - 1
    except ValueError:
        print("Enter a number from 1 to 9.")
        continue
```



```
while True:
    display()
    try:
        move = int(input(
            f"Player {player}, enter position (1-9): "
        )) - 1
    except ValueError:
        print("Enter a number from 1 to 9.")
        continue

    if move < 0 or move > 8 or board[move] != " ":
        print("Invalid move! Try again.")
        continue

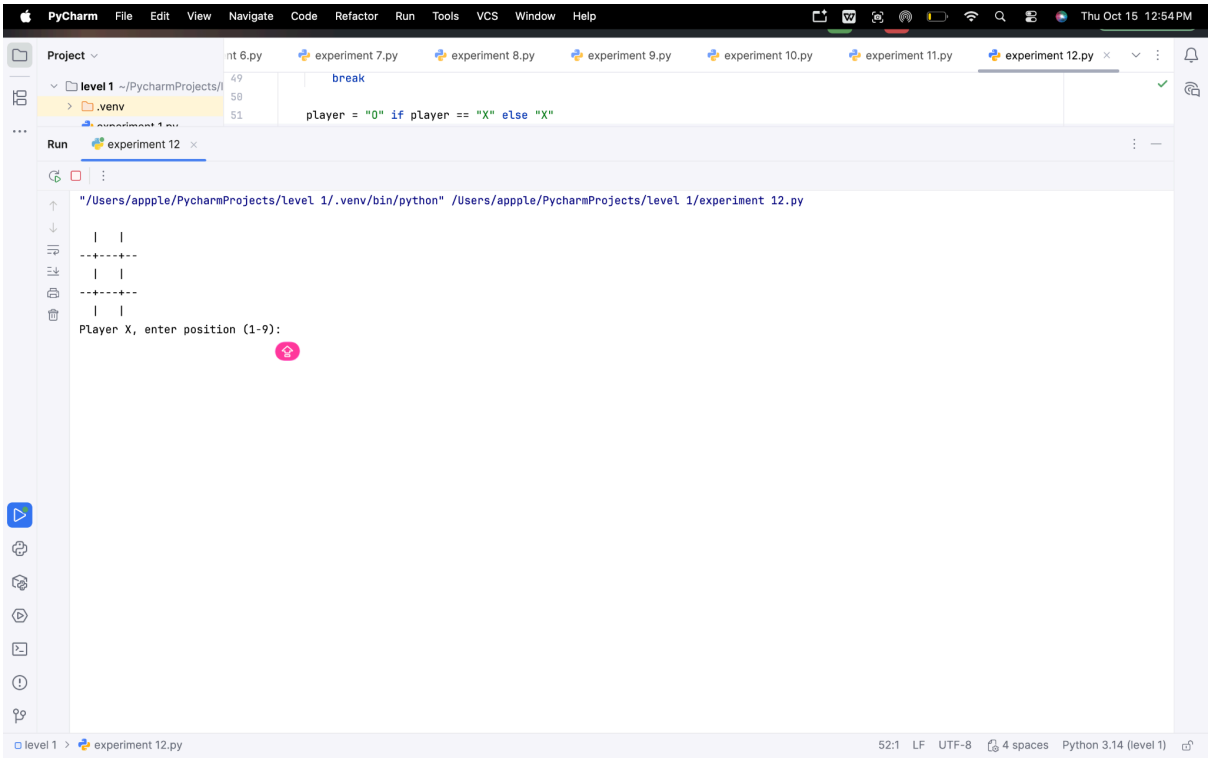
    board[move] = player

    if winner():
        display()
        print("Player", player, "wins!")
        break

    if " " not in board:
        display()
        print("Game Draw!")
        break

    player = "O" if player == "X" else "X"
```

OUTPUT:



Result: Thus, the Tic-Tac-Toe game was implemented successfully using Python.

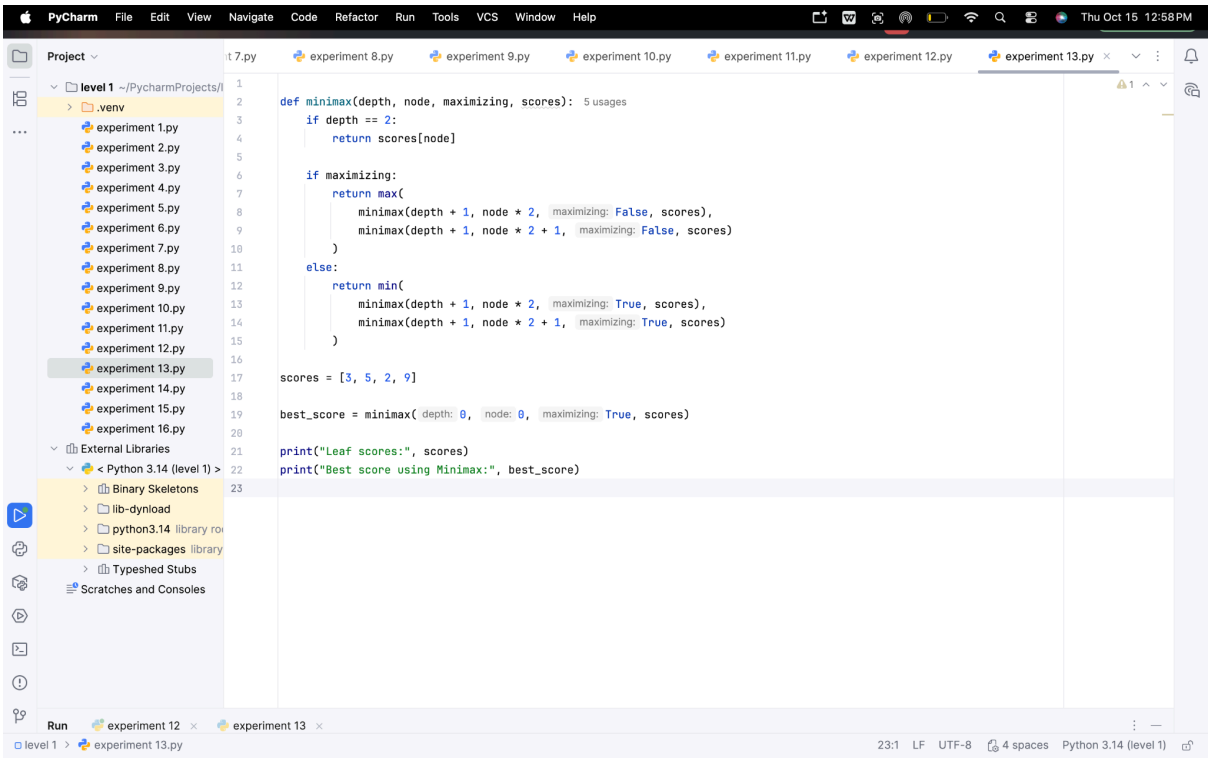
PROGRAM 13: The Python program to implement the Minimax algorithm for gaming

AIM: To implement the Minimax algorithm for selecting the best move in a game.

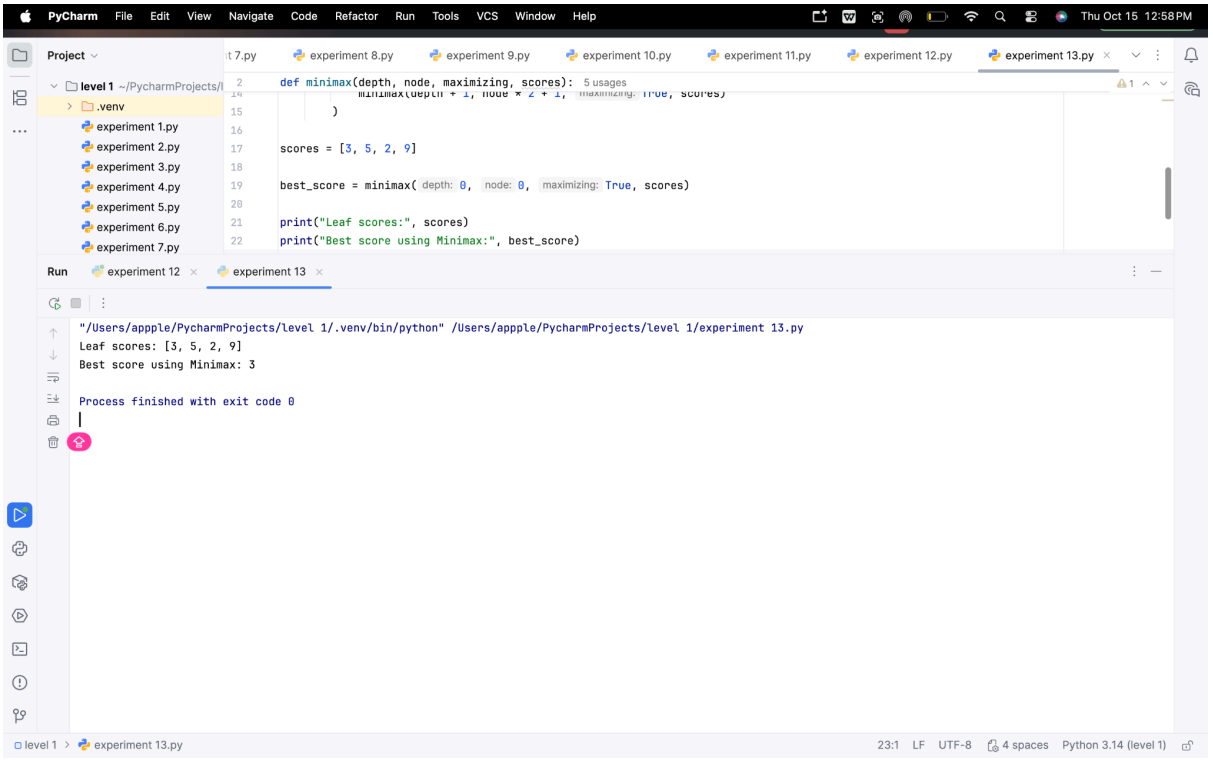
Algorithm:

1. Generate the possible moves.
2. Evaluate the game states.
3. Maximize the score for the computer.
4. Minimize the score for the opponent.
5. Select the move with the best score.

CODE:



OUTPUT:



Result: Thus, the Minimax algorithm was implemented successfully to determine the best score in a game tree.

PROGRAM 14: Python program to implement Alpha & Beta pruning algorithm

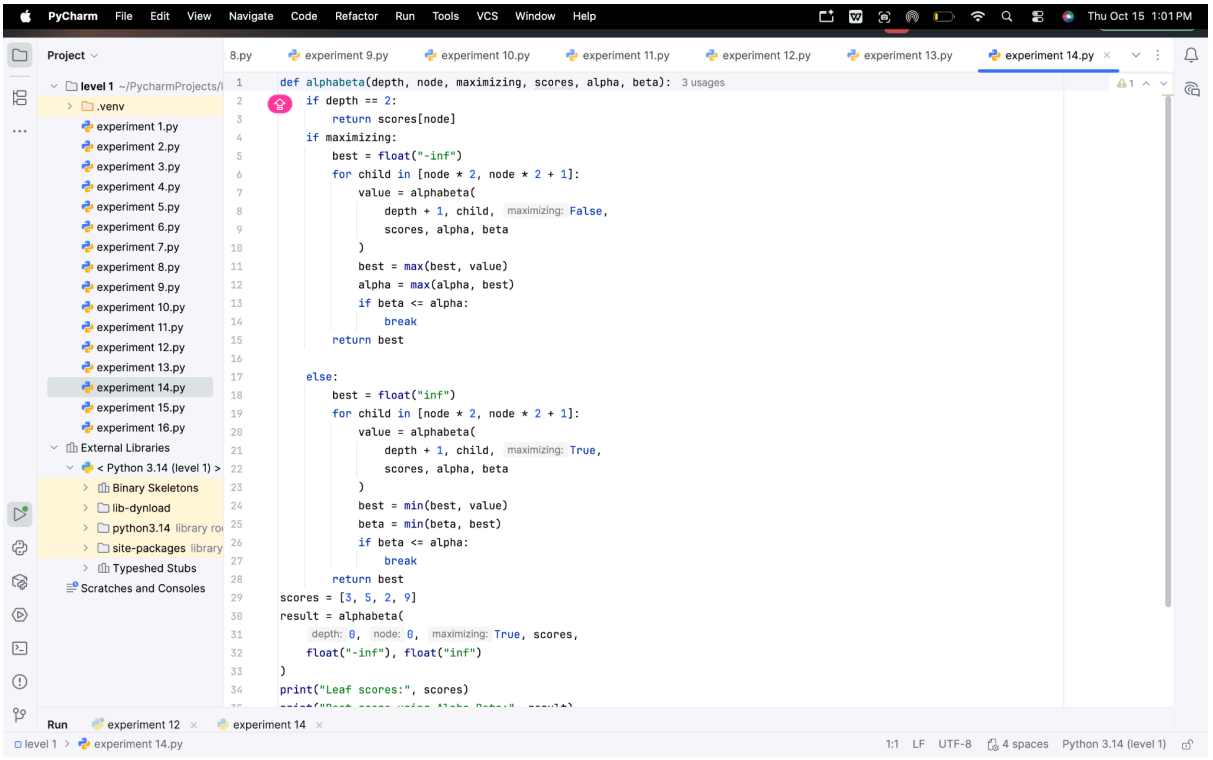
for gaming

AIM: To implement the Alpha-Beta Pruning algorithm to optimize the Minimax search.

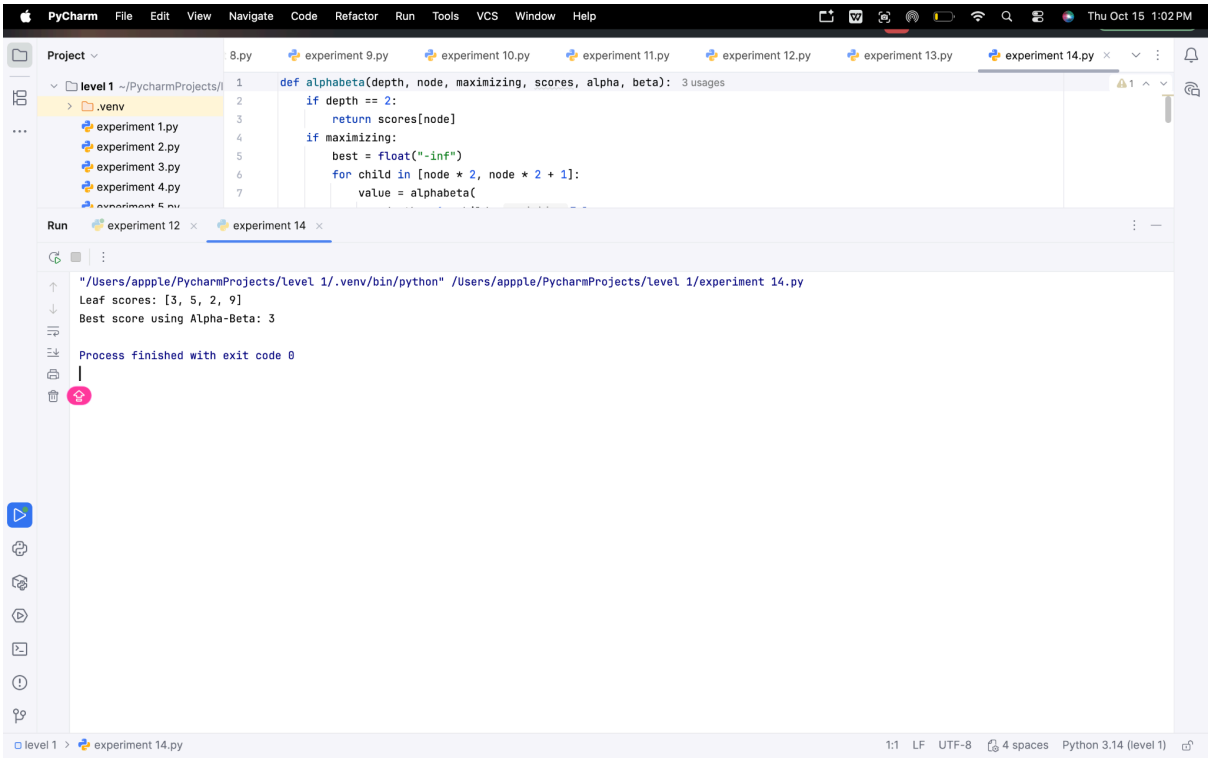
Algorithm:

1. Initialize alpha to negative infinity and beta to positive infinity.
2. Evaluate the game tree recursively.
3. Update alpha at maximizing nodes.
4. Update beta at minimizing nodes.
5. Stop exploring a branch when alpha is greater than or equal to beta.

CODE:



OUTPUT:



Result: Thus, Alpha-Beta Pruning was implemented successfully to reduce unnecessary game-tree exploration while preserving the Minimax result.

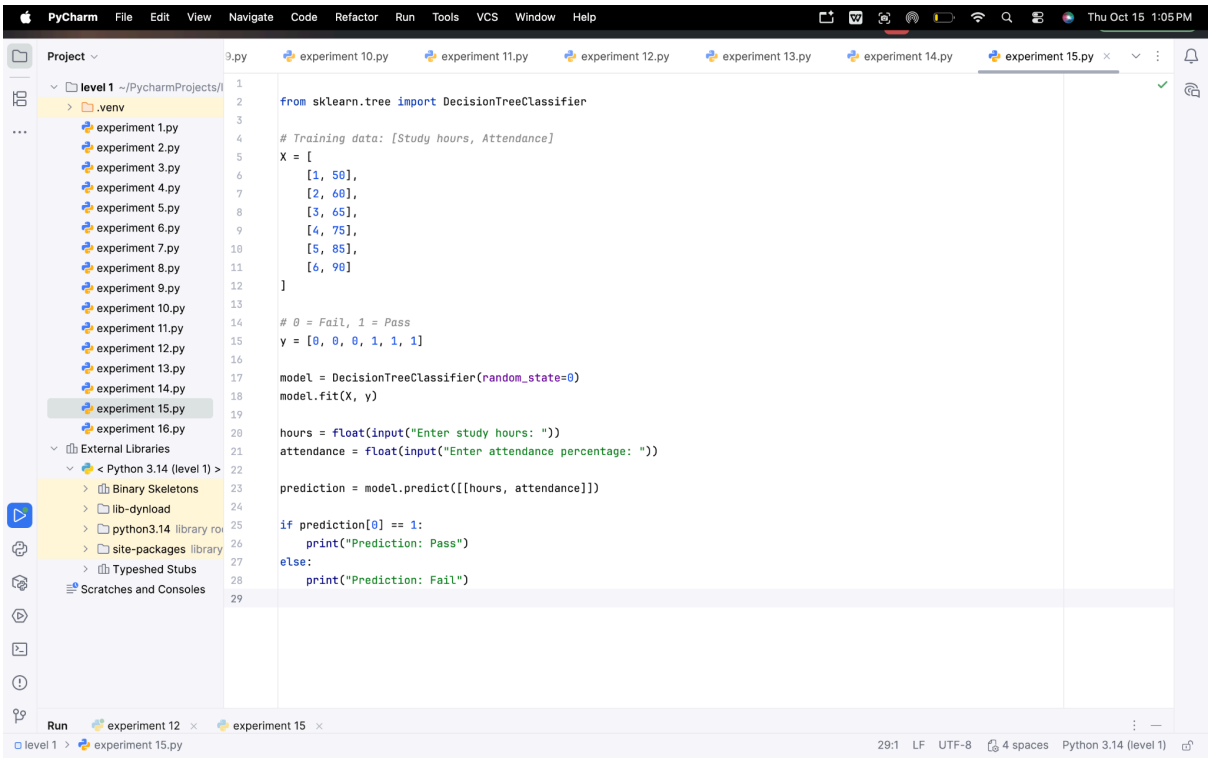
PROGRAM 15: Python program to implement Decision Tree

AIM: To implement a Decision Tree classifier using Python.

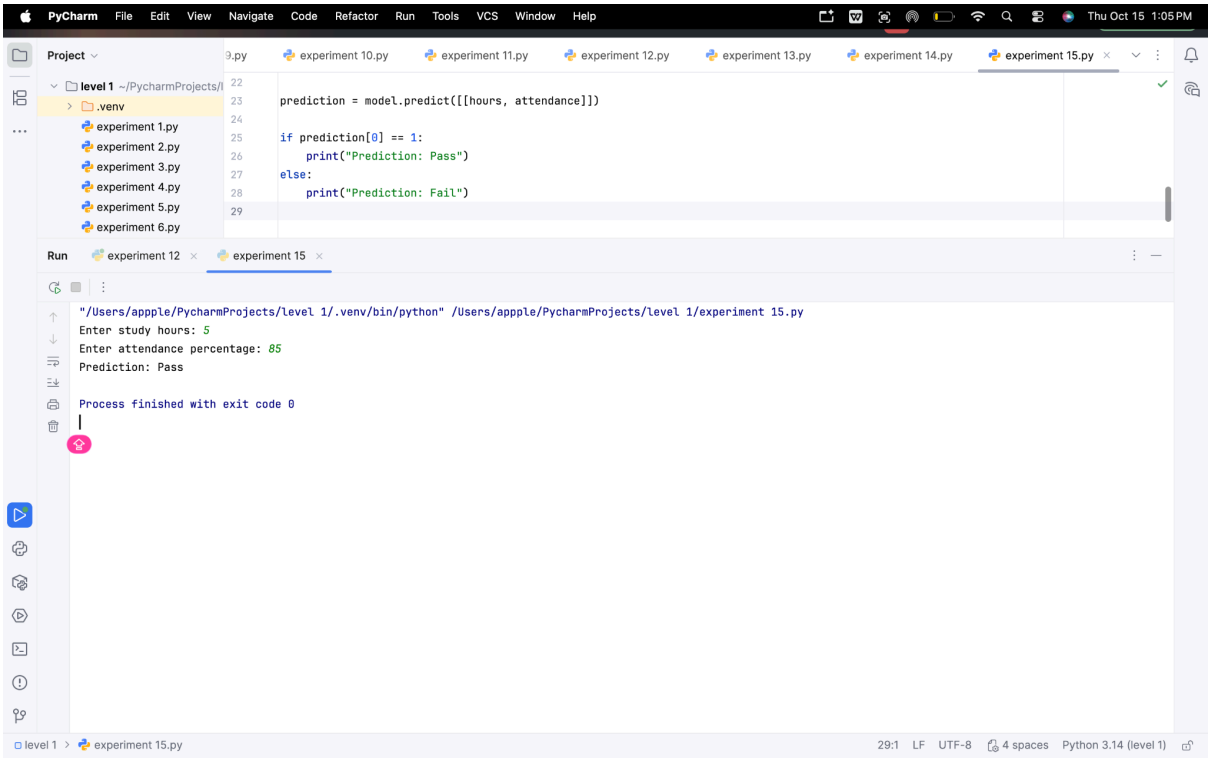
Algorithm:

1. Import the Decision Tree classifier.
2. Prepare the training dataset.
3. Train the classifier.
4. Predict the output for new input data.
5. Display the prediction.

CODE:



OUTPUT:



Result: Thus, the Decision Tree classifier was implemented successfully to predict the result from the training data.

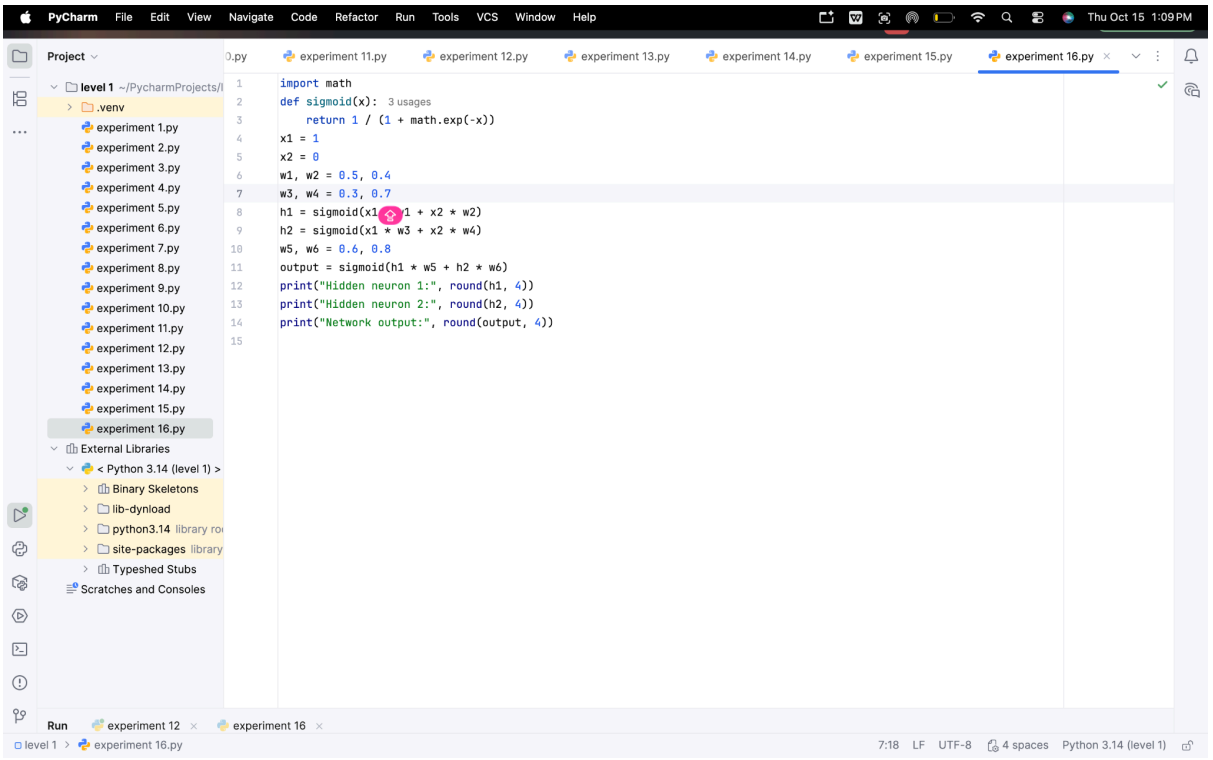
PROGRAM 16: Python program to implement a feedforward neural Network

AIM: To implement a Feed Forward Neural Network using Python.

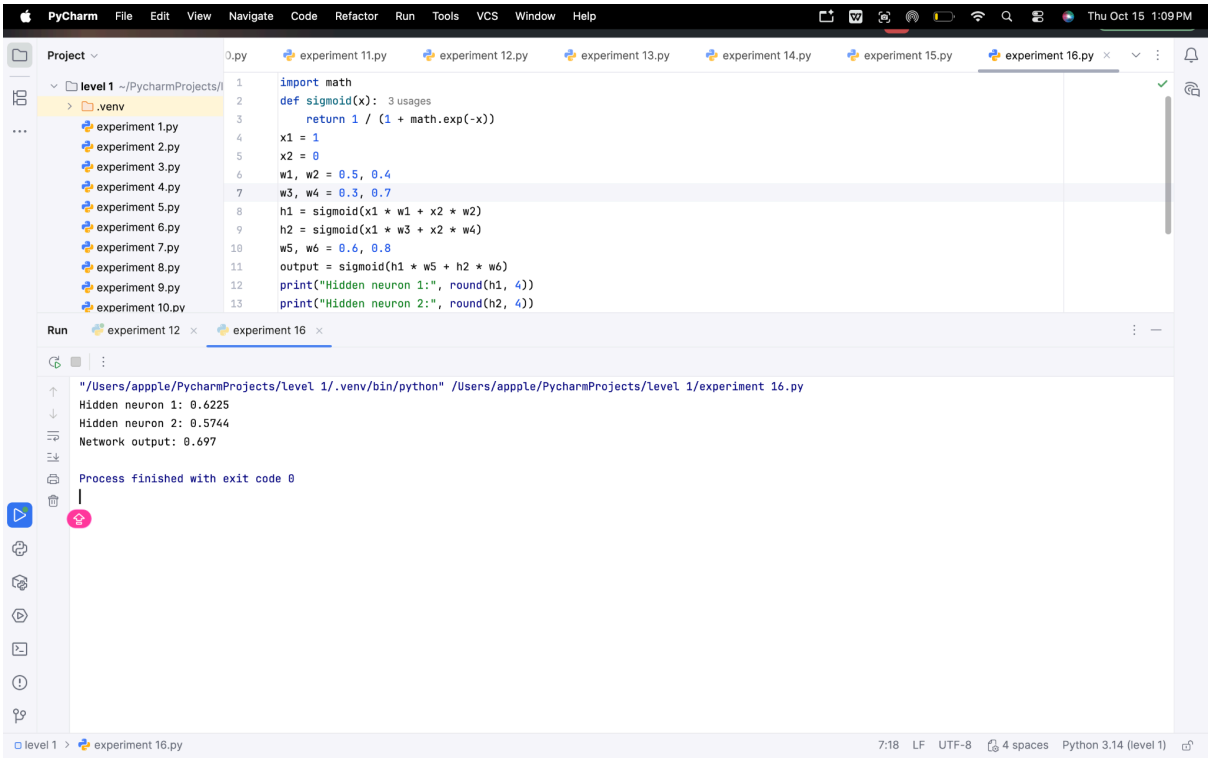
Algorithm:

1. Define the input values and target outputs.
2. Initialize the network weights and biases.
3. Pass inputs through the hidden layer.
4. Calculate the output using the activation function.
5. Display the predicted output.

CODE:



OUTPUT:



Result: Thus, a Feed Forward Neural Network was implemented successfully using forward propagation and the sigmoid activation function.